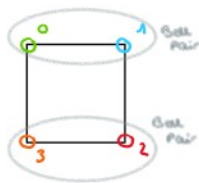


Mean mutual information toy model



coupling_map =

$$= [(0,1), (1,2), (2,3), (3,0)]$$

bell state on (0,1), (2,3)

$$|\Phi^+\rangle = (|00\rangle + |11\rangle) / \sqrt{2} \quad (\mathbb{Z} \text{ basis})$$

$$P(0000) = P(0011) = P(1100) = P(1111) = \frac{1}{4}$$

① Mutual information of a bell pair (actual pair)

→ Joint distribution P_{01}

$$P(q_0, q_1) = (0,0) = P(0000) + P(1100) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

$$P(q_0, q_1) = (1,0) = 0$$

$$P(0,1) = 0$$

$$P(1,1) = 1$$

$$P_{01} = \{00: \frac{1}{2}, 10: 0, 01: 0, 11: \frac{1}{2}\}$$

→ Marginal distribution P_0, P_1

$$P(q_0=0) = P(00) + P(10) = \frac{1}{2} + 0 = \frac{1}{2}$$

$$P(q_0=1) = \frac{1}{2}$$

$$P(q_1=0) = \frac{1}{2}$$

$$P(q_1=1) = \frac{1}{2}$$

→ Shannon entropies: $H(X,Y) = -\sum_{x,y} P(x,y) \log(P(x,y))$

$$H(X) = -\sum_x P(x) \log(P(x))$$

$$\rightarrow \text{pair entropy: } H(q_0, q_1) = -\frac{1}{2} \log \frac{1}{2} - \frac{1}{2} \log \frac{1}{2} - 0 - 0 = 1$$

$$\rightarrow \text{single entropy: } H(q_0) = H(q_1) = -\frac{1}{2} \log \frac{1}{2} - \frac{1}{2} \log \frac{1}{2} = 1$$

→ Mutual information: $MI(X,Y) = H(X) + H(Y) - H(X,Y)$

$$MI(0:1) = H(q_0) + H(q_1) - H(q_0, q_1) = 1 + 1 - 1 = 1$$

② Mutual information of a non-bell pair (single pair)

→ Joint distribution P_{12}

$$P_{12} = \{00: \frac{1}{4}, 01: \frac{1}{4}, 10: \frac{1}{4}, 11: \frac{1}{4}\}$$

→ Marginals: P_1, P_2

$$P(q_1=0) = P(q_1=1) = \frac{1}{2}$$

$$P(q_2=0) = P(q_2=1) = \frac{1}{2}$$

→ Entropies:

$$H(q_1) = H(q_2) = 1$$

$$H(q_1, q_2) = 2$$

→ MI: $I(1:2) = 1 + 1 - 2 = 0$

③ Mean mutual information:

$$\text{mean_mi} = \{ \text{'paired'} : [(1+1)/2] = 1, \text{'single'} : [(0+0)/2] = 0 \}$$

